A Proof of the OEIS A105565 Discrepancy Bound

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Abstract

Let $a(n)$ indicate whether exactly five Fibonacci numbers have n decimal digits, and let $S(n) = \sum_{k=1}^{n} a(k)$. OEIS A105565 conjectures a sharp bounded discrepancy for $S(n)$. We prove the stronger exact identity

$$S(n) = \lfloor n\alpha + \beta \rfloor - 1 \qquad (n \geq 1),$$

where $\alpha = \log(10)/\log(\varphi) - 4$ and $\beta = \log(5)/(2\log(\varphi)) - 1$. The conjectured strict inequalities follow immediately. The proof accounts for the finite cutoff in the Formal Conjectures definition and has been checked by Lean 4 against mathlib without unproved assumptions.

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1 The sequence and the conjecture

Let $(F_k)_{k\geq 0}$ be the Fibonacci sequence with $F_0 = 0$ and $F_1 = 1$. For $n \geq 1$, let $a(n) = 1$ when exactly five values F_k have n decimal digits, and let $a(n) = 0$ otherwise. Write

$$S(n) = \sum_{k=1}^{n} a(k), \qquad \varphi = \frac{1+\sqrt{5}}{2},$$

and define

$$\alpha = \frac{\log 10}{\log \varphi} - 4, \qquad \beta = \frac{\log 5}{2\log \varphi} - 1.$$

This is OEIS A105565, introduced by Reinhard Zumkeller. Hans J. H. Tuenter recorded the discrepancy conjecture in 2025 and also gave a fractional-part description of $a(n)$ for $n > 1$ [1].

Theorem 1. *For every integer $n \geq 1$,*

$$S(n) = \lfloor n\alpha + \beta \rfloor - 1.$$

Consequently,

$$\beta - 2 < S(n) - \alpha n < \beta - 1.$$

The result follows once the exact Fibonacci index cutoff is identified. Two boundary cases matter: no power 10^n is Fibonacci, and the logarithmic cutoff is never an integer. These facts also make the final inequalities strict.

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2 The exact Fibonacci cutoff

For $n \geq 1$, define

$$x_n = \log_\varphi(10^n\sqrt{5}).$$

Binet's formula, with $\psi = (1-\sqrt{5})/2$, gives $F_k = (\varphi^k - \psi^k)/\sqrt{5}$. Since $|\psi| < 1$ and $\sqrt{5} > 2$,

$$\left| F_k - \frac{\varphi^k}{\sqrt{5}} \right| = \frac{|\psi|^k}{\sqrt{5}} < \frac{1}{2}. \tag{1}$$

Lemma 2 (Nonintegral boundary)**.** *For $n \geq 1$, x_n is not an integer.*

Proof. Suppose $x_n = k \in \mathbb{Z}$. Positivity forces $k \geq 1$, and then $\varphi^k = 10^n\sqrt{5}$. The identity

$$\varphi^k = F_k\varphi + F_{k-1}$$

together with $\sqrt{5} = 2\varphi - 1$ yields

$$(2 \cdot 10^n - F_k)\varphi = 10^n + F_{k-1}. \tag{2}$$

The right side is positive, so the coefficient of φ is nonzero. Equation (2) would therefore make φ rational, a contradiction. □

Lemma 3 (Powers of ten are not Fibonacci)**.** *For $n \geq 1$, there is no k with $F_k = 10^n$.*

Proof. If equality held, then $2 \mid F_k$ and $5 \mid F_k$. The identity

$$\gcd(F_r, F_s) = F_{\gcd(r,s)}$$

with $r = 3$ and $r = 5$ implies $3 \mid k$ and $5 \mid k$, respectively. Hence $15 \mid k$, so $F_{15} = 610$ divides F_k. In particular $61 \mid 10^n$. Since 61 is prime, this would imply $61 \mid 10$, which is impossible. □

Lemma 4 (Cutoff equivalence)**.** *For $n \geq 1$ and $k \geq 0$,*

$$F_k < 10^n \iff k < x_n.$$

Proof. The inequality $k < x_n$ is equivalent to $\varphi^k/\sqrt{5} < 10^n$. Equation (1) then gives $F_k < 10^n + 1/2$. Because F_k and 10^n are integers, $F_k \leq 10^n$; equality is excluded by Lemma 3.

Conversely, suppose $F_k < 10^n$. If $k \geq x_n$, then Lemma 2 gives $k > x_n$, so $\varphi^k/\sqrt{5} > 10^n$. Equation (1) would imply $F_k > 10^n - 1/2$, contradicting the integrality of F_k and the strict upper bound $F_k < 10^n$. □

Set $c_n = \lfloor x_n \rfloor$. Lemmas 2 and 4 show that

$$F_k < 10^n \iff k \leq c_n. \tag{3}$$

3 Reduction to a mechanical word

Expanding the logarithm gives

$$x_n = 4n + 1 + (n\alpha + \beta). \tag{4}$$

Put $q_n = n\alpha + \beta$ and $b_n = \lfloor q_n \rfloor$. Then $c_n = 4n + 1 + b_n$.

The elementary bounds $\varphi^4 < 10 < \varphi^5$ give

$$0 < \alpha < 1. \tag{5}$$

Similarly, $\varphi^6 < 10\sqrt{5} < \varphi^7$, combined with $x_1 = 5 + \alpha + \beta$, gives

$$1 < \alpha + \beta < 2. \tag{6}$$

In particular $q_n > 0$. Equations (5) and (6) also give $q_n < n + 2$, so $b_n < n + 2$ and

$$c_n < 5n + 10. \tag{7}$$

This verifies that the finite search range $[0, 5n + 10)$ in the Formal Conjectures definition contains every relevant Fibonacci index. The formal theorem therefore proves the repository definition itself, not merely an unbounded reformulation.

For $n \geq 2$, equation (3) shows that the indices of Fibonacci numbers with exactly n digits are precisely the integers in $(c_{n-1}, c_n]$. Their number is

$$c_n - c_{n-1} = 4 + (b_n - b_{n-1}). \tag{8}$$

Because $0 < \alpha < 1$, the floor increment $d_n = b_n - b_{n-1}$ is either 0 or 1. Equation (8) shows that exactly five n-digit Fibonacci numbers occur exactly when $d_n = 1$. Hence

$$a(n) = b_n - b_{n-1} \qquad (n \geq 2). \tag{9}$$

Proof of Theorem 1. The initial value is $a(1) = 0$. By equation (6), $b_1 = \lfloor \alpha + \beta \rfloor = 1$. Summing equation (9) therefore telescopes:

$$S(n) = b_n - 1 = \lfloor n\alpha + \beta \rfloor - 1.$$

Lemma 2 and equation (4) show that q_n is never an integer. Consequently $q_n - 1 < b_n < q_n$. Substituting $q_n = n\alpha + \beta$ and $S(n) = b_n - 1$ gives

$$\beta - 2 < S(n) - \alpha n < \beta - 1,$$

as required. □

4 Machine-checked verification

The argument was formalized in Lean 4 against the Google DeepMind Formal Conjectures repository [3, 4]. The formalization proves the exact bounded-filter sequence definition and the theorem `OeisA105565.conjecture`. It was checked against repository commit

`2411d22e1bd550d050d0eac6c1fb379a76a3e7c5`

using Lean/mathlib 4.27.0. The following checks completed successfully.

Check	Result
File	`lake env lean FormalConjectures/OEIS/105565.lean` passed.
Module	`lake -wfail build FormalConjectures.OEIS.105565` completed 8,055 jobs successfully.
Axioms	`propext`, `Classical.choice`, and `Quot.sound`; no `sorryAx`.
Sanity	No `sorry`, `admit`, `axiom`, or `unsafe` declaration occurs in the target file.

A kernel check does not replace editorial review of exposition or novelty, but it certifies that the theorem follows from the displayed Lean definitions and mathlib's accepted axioms [5, 6].

Contribution and AI-use statement

Adam Haig supplied the prompt, selected the conjecture and acceptance criteria, authorized publication, and reviewed the resulting mathematical and formal artifacts. OpenAI Codex, an AI coding and reasoning system, performed the majority of the mathematical work: it developed and refined the cutoff argument, located relevant mathlib lemmas, generated and iteratively repaired the Lean formalization, executed the verification checks reported above, and drafted this note. The byline credits both contributors while identifying Codex explicitly as an AI system. Haig is the human submitting author and accepts responsibility for the deposit and for correcting any error that may be found.

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